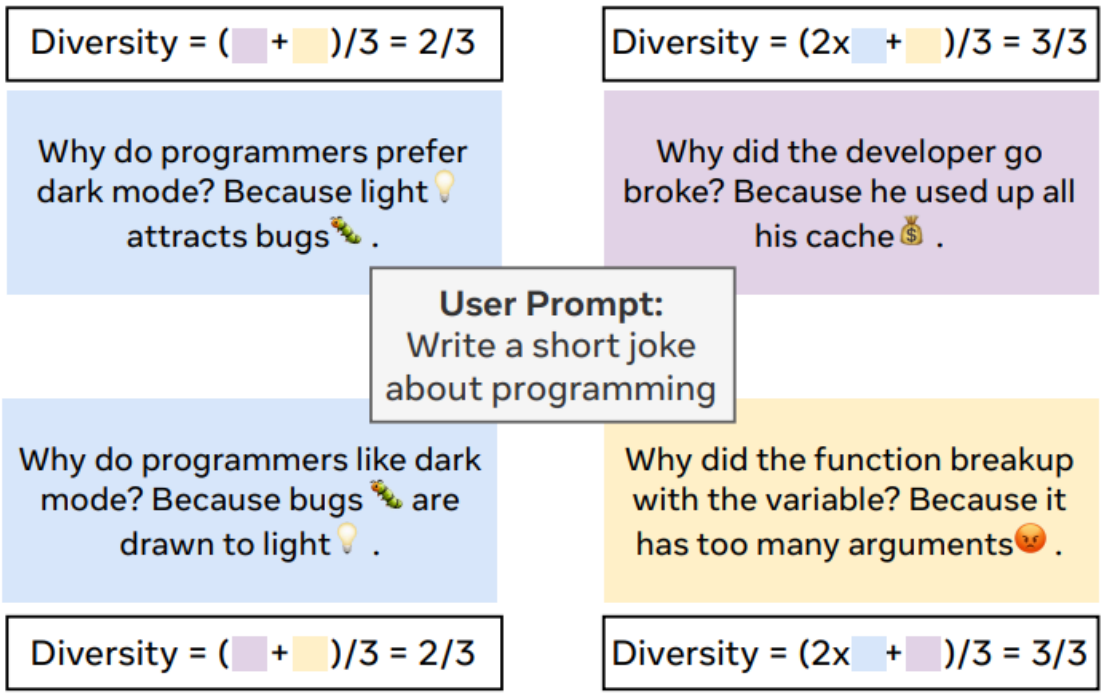
Darling: Jointly Reinforcing Diversity and Quality in Language Models

> #RL #Inference #GRPO #Diversity

- What problem does this framework solve?
  - Post-training, such as RL and SFT, can significantly improve the quality of the response of the model.
  - However, it is often observed that though the quality of answers improves, the variety decreases.
  - **D**iversity-**A**ware **R**einforcement **L**earning (DARLING) is an RL framework to keep both variety and quality.
- What does this framework do?
  - This framework is based on the classic GRPO algorithm.
  - Unlike the classic GRPO, which is solely based on the reward that measures the *quality* of the answer, DARLING also measures semantic diversity.
  - To achieve this, a classifier is trained, formally:

    $$\mathbf{classify}(y_i, y_j) = \mathbf{1}(y_i \text{ semantically equivalent to } y_j)$$
  - For each prompt, we generate $n$responses, and calculate the total score of the group by pair (we negate the $\mathbf{classify}(\cdot,\cdot)$to give 1 to different pairs):

    $$\mathrm{Div}_d(y_i \mid y_1, \ldots, y_n) = \frac{1}{n-1} \sum_{\substack{j=1 \\ j \neq i}}^{n} \neg \mathbf{classify}(y_i, y_j).$$
    - e.g.,



  - We combine the diversity score and the quality score by multiplying them:

    $$r_{\text{darling}}(x, y_i \mid y_1, \ldots, y_n) := r(x, y_i) \times \mathrm{Norm}(\mathrm{Div}_d(y_i \mid y_1, \ldots, y_n))$$
    - *We don't add them together as there will be a way of cheating: to achieve a high score, the model can either generate low-diversity answers but with very high quality or bad answers with high diversity.*
  - Use the $r_{darling}$to calculate the advantage:

$$A_{i,t} = r_{\text{darling}}(x, y_i \mid y_1, \ldots, y_n) - \text{mean}_{j=1}^{n}(r_{\text{darling}}(x, y_j \mid y_1, \ldots, y_n)).$$

        - In this case, it is found that without dividing the std will give a more stable convergence.
    - Got high scores in both non-verifiable and verfiable (like math) benchmarks.
- My Thought
    - **Pairwise semantic diversity calculation optimization**: current operation has a time complexity of $O(n^2)$. It is possible to optimize this by using Distance to Centroid as the reward. Plus, Distance to Centroid is non-binary, so the reward can be more dynamic.
    - **Expand the usage to CoT**: instead of working on output, can we use the same framework to evaluate and optimize the chain of thought, meaning we give the DARLING reward to `<think>...</think>`. Plus, we need to make sure the final output is correct, meaning that we need to modify the reward function to:

      $$r_{\text{darling CoT}}(x, \text{CoT}_i \mid \text{CoT}_1, \ldots, \text{CoT}_n, y_i) := r(x, \text{CoT}_i) \times \text{Norm}(\text{Div}_d(\text{CoT}_i \mid \text{CoT}_1, \ldots, \text{CoT}_n)) \times \mathbb{I}(y_i = y_i^*)$$
- Work Cited
    - Paper
    - Chat History